

Survey on Agile and Lean Usage in Finnish Software Industry

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ABSTRACT

Earlier empirical studies have demonstrated the interest that agile methods have generated in the software industry. Currently, lean approaches are increasingly adopted for complementing agile methods in software processes. With the goal of providing up-to-day results that can be used by organizations implementing or planning to implement agile and/or lean methods, we have conducted a study on the current stage of agile and lean adoption and usage in the software industry. For this purpose, we conducted an extensive survey among Finnish software practitioners in 2011, using the membership registry of *The Finnish Information Processing Association* (FIPA) as a sampling frame. 408 responses were collected from 200 software intensive organizations in the study. The survey included questions for identifying the rate of agile and lean usage in software organizations as well as the implementation of specific methods and practices, goals in adopting agile and lean, reasons for not applying these methods and effects of the agile and lean usage. The results of the survey reveal that a majority of respondents' organizational units are using agile and/or lean methods (58%). Furthermore, lean appears as a new player, being used by 24% of respondents, mainly in combination with agile (21%). The reasons and benefits for using agile and lean methods appeared to correspond in most parts to the findings of the earlier research. Generally, the experiences of using agile and lean methods seem to be rather positive, although challenges, such as obtaining management support and limitations for scaling agile in distributed settings, were also identified.

Categories and Subject Descriptors

D.2.9 [Software Engineering]: Management – *life cycle, productivity, programming teams, software process models.*

D.2.10 [Software Engineering]: Design – *Methodologies.*

A.1 [Introductory and Survey].

General Terms: Management, Design, Economics, Human Factors.

Keywords: Agile, lean, survey.

1. INTRODUCTION

Agile Software Development (ASD) can be seen as a reaction against traditional plan-driven methodologies, which were considered unable to meet the dynamism, unpredictability and changing conditions that characterize the current business environment for software intensive organizations [1]. Since the

Agile Manifesto¹ was published in 2001, the feedback on ASD has been relatively positive [2, 3, 4, 5, 6]. However, challenges such as how to scale ASD to large contexts have also been identified [7, 8, 9].

In more recent years, and mainly motivated by the agile community, the software industry has started to look at lean [10] as a new approach that could complement ASD. Lean Software Development (LSD) is aligned with many agile principles but considering a more holistic and enterprise system thinking [11]. The discussion, started as early as the 1990s [12], about the application of lean for software development has become again a trend. Thus, while progress towards LSD is mainly driven by industry pioneers, there is a growing body of literature documenting case studies [13, 14, 15] and investigating specific elements of lean in software development, e.g., flow [16, 17].

The universal application of the core lean principles in a creative activity as software development is under debate [15]. Furthermore, the relationship between ASD and LSD has not been clearly defined [18], limiting the current comprehension of the phenomenon. Nevertheless, we are interested in the fact that software organizations are moving towards agile and lean in the real-world industry context and in what we can learn from their experiences. In the past years, with a few exceptions (such as [8, 19, 20, 21], most of scientific research on ASD has been based on case studies and experimental works [1]. It is clearly recognized that these works provide detailed and valuable analysis of the usage of ASD for specific projects. However, they are limited for drawing general conclusions. Consequently, while the Agile Manifesto was formulated more than 10 years ago, little knowledge has been generalized about the usage of agile methods and its benefits and problems [1].

With the goal of contributing to provide more generalizable findings, and following the indications of studies analyzing the research agenda for ASD [1, 22], we have conducted a large-scale exploratory survey study for addressing the following questions: (a) What is the current state of adoption and usage of ASD and LSD methods and practices in the software industry? (b) What are the reasons why ASD and LSD are being adopted in some software development organizations? (c) Which are the impacts, in terms of benefits, of using ASD and LSD? (d) Which are the limitations and factors that can challenge the usage of ASD and LSD? and (e) Which are the reasons of some organizations for not using ASD and LSD?

The survey study, conducted in the Finnish software industry in 2011, collected more than four hundred responses of software practitioners. Large number of practitioners was from multinational companies with premises in several countries. Furthermore, the results of the study are especially interesting if it

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¹ <http://www.agilemanifesto.org/>

is considered that Finland takes up the second position in the IT Industry Competitiveness Index 2011 of the BSA/Economist's report [23]. Therefore, although the generalization of the results is still limited, we consider the Finnish software industry as a suitable population for the study.

The paper makes the following contributions to research and practice: i) it provides more generalizable and up-to-date results on the state of ASD and LSD usage, using first-hand industrial insight on how ASD and LSD are being used in the real-world industry. The findings may be advantageously leveraged by other organizations considering whether/implementing agile and/or lean, through a better understanding of the methods and practices that their peers are using as well as their benefits and limitations in using ASD and/or LSD. ii) to our best knowledge, this study presents the first large scale explorative study analyzing the status of lean usage for software development, iii) moreover, the results of the study provide basis for future research by identifying the most applied methods and practices as well as the most experienced benefits and challenges on using ASD and LSD. These elements should guide the research agenda of other initiatives, such as case studies or experiments, which study in deeper detail the specific problems that organizations face when deploying ASD and LSD.

The rest of the paper is organized as follows: Section 2 reviews earlier studies on agile and lean usage. Next, section 3 describes the research setting, including the process for collecting the data as well as the design of the survey. Section 4 includes a systematic and extensive presentation of the survey questions and survey results. Finally, section 5 concludes the paper summarizing the results and presenting the limitations of the study.

2. EARLIER STUDIES ON AGILE AND LEAN USAGE

Most of earlier surveys on agile usage have been conducted by IT consultants. Although the results of this kind of studies can be used for guidance, their consideration as scientific contributions is questionable.

The last “*State of agile development*” survey [24] conducted by Version One between July and November of 2011 received 6042 responses (rate of response and design of the sample were not provided). The survey's findings showed that more than 80% of respondents' companies had adopted agile in some extent and nearly half of respondents indicated that their companies had practiced agile for over 2 years. Scrum was indicated as the most followed agile method (53% respondents were following it), followed by a hybrid usage of Scrum and Extreme Programming (14%). Only 2% of respondents indicated to be using lean. Regarding to agile practices, the survey found daily stand-up (78%), iteration planning (74%) and unit testing (70%) as the top followed practices. The top three reasons cited for adopting agile were to accelerate time to market, to increase productivity and to more easily manage changing priorities. Benefits that were actually experienced according to the respondents of this survey were the ability to manage changing priorities (84%) and improved project visibility (77%). Finally, the findings of the survey indicated the inability to change the organization's culture as the biggest barrier to agile adoption (52%), followed to the availability of personnel with right skills (40%). However, these results must be cautiously considered since Version One is a vendor of products and training on agile methods and, therefore, has vested interests in the results of the survey they conduct.

The Scott Ambler's February 2008 Agile Adoption Survey [25] collected 642 responses. The sample of the study was composed mainly by IT professionals in North America who were familiar with agile methods. 69% of respondents indicated that they were working in organizations that had some experience with agile. The results of the survey also indicated that collocated projects were more successful than no-collocated ones when using agile methods (82% and 71% rate of success respectively). Also, the respondents of the survey indicated that they had achieved benefits in productivity (88% of respondents answered somewhat higher or much higher productivity), quality (77% of respondents indicated somewhat higher or much higher quality of systems deployed) and business stakeholder satisfaction (78% of respondents indicated somewhat higher or much higher business stakeholder satisfaction).

A similar survey was conducted by Forrester Research from July to August 2009 [26]. The survey was fielded by 1298 application development and program management professionals who were readers of Dr. Dobbs's magazine². In this survey, 35% of respondents stated that agile most closely reflects their development process. However, 34% indicated that they continue to use either an iterative or waterfall process as their primary software development method. Regarding to the agile practices implemented, short iterations (79%), constant feedback (77%) and daily scrum meeting (71%) were the most implemented practices between the development professionals who had adopted agile. Oppositely, test-driven development (TDD) and system metaphor were the least implemented practices (42% and 15% respectively).

Regarding to studies published in scientific forums such as conferences and journals, most of them have focused on analyzing in detail specific elements in the usage of ASD, using research methods such as case studies [1]. Just as an example, Pikkariainen et al. [4] studied the impact of agile practices in communication through a case study in F-Secure. The results of the study indicated that agile practices had a positive impact on both communication inside development teams and communication outside team boundaries. However, it was found that the communication mechanisms that ASD provides for “*extended environments*” (with many stakeholder groups) are insufficient.

Some scientific studies have analyzed important factors in adopting and using ASD. Chow and Cao [19] studied critical success factors in agile software projects. Based on a confirmatory survey study, collecting data from 109 agile projects from 25 countries, Chow and Cao identified the following success factors: agile-friendly project team environment, high-caliber team capability, strong customer involvement, agile style project management process, agile style software engineering techniques and correct delivery strategy. Similarly, Misra et al. [21] carried out a survey-based ex-post facto study for identifying those factors that impact in the success of projects developed using ASD. The study, based on 241 responses, found personal characteristics of team members, society in which the organization operates, training and learning, customer satisfaction, collaboration and commitment, fast decision making, corporate culture and self-team control as important success factors. Other factors such as communication and negotiation capabilities or team distribution were not found as statistically significant. Contrarily, Hanssen et al. [9], using a tertiary review of the literature, found that although ASD is a more and more stable trend in global software engineering, the use of ASD in globally

² <http://www.drdobbs.com/>

distributed environments may have limitations. For example, lack of effective collaboration tools or the influence of cultural and linguistic diversity in communication and collaboration were seen as possible challenges. Thus, Hanssen et al. concluded that while some challenges seem solvable, other agile practices cannot be applied in a distributed setting.

Only few scientific studies have explored the level of agile adoption from an extensive perspective. Salo and Abrahamsson [27] investigated the use and usefulness of Extreme Programming (XP) and Scrum in European embedded software development organizations. The study is based on the data collected from a sample of 13 industrial organizations (from eight European countries) and 35 individual projects. Open office space (66% of systematically or mostly used responses), coding standards (60% of systematically or mostly used responses) and 40h week-sustainable pace (59% of systematically or mostly used responses) were reported as the most used XP practices. In addition, TDD and pair programming were found as the least “systematically or mostly” used XP practices. Regarding to Scrum, product backlogs and daily scrum meetings were the most used practices (24% and 21% of systematically or mostly used responses respectively). A significant finding of the study was the high percentage of “never” or “I don’t know” responses that all Scrum practices received in this study. The authors interpreted this result as unfamiliarity with Scrum among a large proportion of respondents.

Vijayarathy and Turk [8] conducted a survey study to assess the factors influencing agile adoption as well as benefits, challenges and limitations of agile development in 2008. The survey study, conducted among fifteen Yahoo online discussion groups on ASD, received 98 responses. Project turn-around time, complexity of software and stability of requirements were reported as the main drivers in organizational decisions to use agile. Regarding to benefits of using agile, better meet customer needs, improved software quality and increased flexibility in development were found as the top benefits. Also, the results of the study showed organizational resistance and management apathy as the main challenges in using agile, and limited support for distributed development environments and limited support for development involving large teams as limitations of agile methods.

Asnawi et al. [28] have recently investigated on agile methods usage in Malaysia. By interviewing 13 software development practitioners from seven organizations, they found that social and human aspects are more important than technical aspects when using agile methods.

With similar objectives than this study, Begel and Nagappan [7] presented the results of a survey study conducted at Microsoft in ESEM 2007. The goals of the study were to assess the extent to groups at Microsoft Research were using ASD and the benefits and problems associated with its usage. They found that 32% of the respondents were using ASD, being Scrum the most used method with 65% of respondents using it. Team coding standards and continuous integration were found as the top agile practices that teams followed, while pair programming and TDD were the least followed practices. Regarding to the benefits and problems associated to the usage of ASD at Microsoft Research, improved communication and coordination as well as the capacity to deliver quicker releases appeared as the most important perceived benefits. On the other hand, whether ASD scales to larger software teams and the inefficiency of too many meetings of methods such as Scrum were seen as the main problems. Although the results of this study provide valuable insight, the study is based on only one organization and its results cannot be

generalized. Thus, Begel and Nagappan requested collaboration for replicating the study in a larger scale to provide more generalized knowledge on the status of ASD in software development organizations. Table 1 synthesizes the most important findings of previous studies regarding to ASD usage.

Table 1. Synthesis of previous studies on ASD usage

<i>ASD usage rate</i>	• Version One [24]: 80% • Scott Ambler [25]: 69% • Forrester Research [26]: 35%
<i>Most used agile practices</i>	• Daily stand-up [24, 26, 27] • Iterative development [24, 26] • Coding standards [7, 27] • Continuous Integration [7], unit testing [24], constant customer feedback [26], 40 hours week [27], product backlog [27], open office space [27]
<i>Least used agile practices</i>	• TDD [7, 26, 27] • Pair programming [7, 27] • System metaphor [26]
<i>Benefits</i>	• Flexibility, ability to manage changing priorities [8, 24] • Improved software quality [8, 25] • Improved project visibility [24], improved productivity [25], reduced time-to-market [7], improved communication and cooperation [7], improved stakeholders -customer satisfaction [25], better meet customer needs [8]
<i>Challenges and limitations</i>	• Difficulties for scaling agile in the large [7, 8, 9, 25] • Difficulties for changing the organizational/people culture [8, 24] • Difficulties for finding personnel with right skills [24], possible inefficiency of too many meetings in methods such as Scrum [7]

The research activity of LSD, viewed originally as just another agile method, is still an emerging area [18]. Thus, studies focused on LSD are quite scarce [13, 14, 15, 16, 17]. Although dedicated conferences and magazines, such as Leanssc, Less and Lean Magazine³, are emerging and contributing to increase the body of literature in the topic, there are not studies, at least at public level, that have analyzed the level of LSD usage.

As philosophies, lean and agile have different origins and some intertwined ideas. Agile focuses more on flexibility and the capacity to rapidly embrace change. While lean focuses on overall economic contribution from a more holistic enterprise perspective [29]. In the field of software engineering, discussions about software development and lean thinking started as early as the 1990s [12], well before the *Agile Manifesto* was formulated. Moreover, lean has been considered as one of the inspiring sources of the *Agile Manifesto* [30]. Therefore, it is reasonable that there exists a significant overlap and some confusion between agile and lean in software development, since both paradigms share numerous elements. Poppendieck considers lean thinking a “platform upon which to build agile software development

³ Leanssc: Lean Software and Systems Conference, <http://www.leanssc.org/>. Less: Conference on Lean Enterprise Software and Systems, <http://less2011.leanssc.org/>. Lean Magazine, <http://leanmagazine.net/>

practices.” [31]. Coplien and Bjornwig argue that although agile and lean have fundamental differences, yet complement each other by addressing different components of systems development [32]. As our approach in this study was essentially empirical, conceptual analysis and refinement of formal definition of agility or leanness, and their relationship, theoretically for software development was beyond the scope of this work. We were interested in the fact that software companies transitioning toward lean usually incorporate elements of agile and we wanted to empirically analyze how this phenomenon is happening and experienced by the organisations in the real-world software industry, in order to identify most important elements that impact their usage.

3. RESEARCH SETTING

The agile and lean usage survey was conducted among Finnish software professionals in June 2011, using Webropol Internet survey tool. In this section, we present the research setting of the survey, describing the data collection and survey design.

3.1 Data Collection

The target population of the survey was Finnish software practitioners. In Finland, there exists an independent association of Finnish ICT professionals and companies called *The Finnish Information Processing Association* (FIPA)⁴. It has about 16 000 professionals as personal members and the number of company members is more than 500. The membership registry of FIPA provided an excellent sampling frame for the survey, as large part of software professionals in Finland are members of it. FIPA provided from the membership registry a selection of 4450 e-mail addresses of software practitioners that suited to the focus of the survey. After piloting the questionnaire for checking its consistency and legibility, the Internet survey request was e-mailed to these persons and the survey was open for two weeks. During this time, 408 persons responded to the survey. These respondents represented 200 different organizations. The response rate was 9%.

3.2 Survey Design

The survey was designed to describe the extent of usage of agile and lean methods, principles and practices within the software practitioner’s organization as well as to explore the reasons for adopting agile and lean, and effects in the adoption. The survey, including almost fifty questions, was designed based on the literature on agile and lean as well as earlier empirical studies of agile and lean adoption and usage. The survey questions consisted of the following sections: background information; usage of agile and lean methods; usage of specific methods, practices and principles; goals, challenges and effects in adoption; non-adopter’s reasons and plans to adopt agile and lean methods.

Background information of the practitioners and their organization consisted of the following questions: position in the organization, years of experience in software development, name of the organization, number of employees in the organization, number of employees in the organizational unit, typical team size.

Usage of agile and lean methods consisted of the following questions: usage of agile and lean methods, duration of using agile and lean methods in the organizational unit, level of adoption of agile and lean methods in the organizational unit.

Usage of specific methods, practices and principles consisted of the following questions: usage of specific agile and lean methods, usage of specific agile and lean practices, usage of specific agile and lean principles.

Goals, challenges and effects in adoption consisted of the following questions: goals in agile and lean adoption, challenges in agile and lean adoption, limitations affecting agile and lean adoption, effects of adopting agile and lean in the organization.

Non-adopters reasons and plans to adopt agile and lean methods consisted of the following questions: reasons for not adopting agile and lean, plans to adopt agile and lean in the future, agile and lean methods planned to be adopted, reasons for adopt agile and lean in the future.

Although preliminary designed options were presented in the last three sections, the survey included open fields for indicating other options not included in the preliminary list.

4. RESULTS

This section presents the results of the survey. A preliminary analysis of companies’ goals driving towards the adoption of agile and/or lean methods has previously been reported in [33]. The results are organized according to the sections of the survey and compared with the results of earlier studies on agile and lean usage.

4.1 Background Information

The number of practitioners answering the survey was 408. The respondents were working in various positions in their organizations. The main organizational roles of the respondents were developers (n=113) and project managers (n=99). Table 2 and Table 3 present the positions of the respondents in their organizations and the respondent’s experiences in software development respectively.

Table 2. Positions in organization

Position ⁵	n	Position	n
Developer	113	Scrum master	33
Project manager	99	Process manager	31
IT staff	79	Product owner	25
Architect	63	Product manager	23
Consultant/Trainer	52	President/VP/CEO/COO/CIO/CTO	22
Quality assurance/Tester	38	Sales/Marketing personnel	10
Operations/Support staff	35	Other	48

Table 3. Experience in software development

Years of experience	n	%
None	42	10,3
Less than 2	31	7,6
2-5	56	13,7
5-10	80	19,6
10-20	144	35,3
More than 20	55	13,5
Total	408	100,0

⁴ <http://www.ttlry.fi/english>

⁵ Since the same person can hold different positions, respondents were able to select multiple options

The respondents belonged to 200 different organizations. The majority of the organizations were companies, but there were also a few educational and governmental organizations. Some of the respondents (n=30) did not identify their organizations. Most of the organizations were big (46%, more than 1,000 employees) and middle size (45%, number of employees between 11-1000). The small organizations with 10 or less employees had 8% representation in the data. For one percent of the respondents the organization size information was missing.

The average size of the respondent's organizational units is presented in Table 4. In most of the organizations, the organizational unit size appears to be rather small, and three quarters of the organizational units include less than 100 persons.

Table 4. Size of the organizational unit

Employees	n	%	Cumulative %
1-10	93	23,8	23,4
11-50	141	35,4	58,8
51-100	66	16,6	75,4
101-200	44	11,1	86,4
201-500	28	7,0	93,5
501-1000	16	4,0	97,5
More than 1000	10	2,5	100,0
Total	398	100,0	

The typical size of working teams in the respondents organizational units are presented in Table 5 below. For ten percent of the respondents the team size information was missing.

Table 5. Team size

Team size	Frequency	%	Cumulative %
1-5	156	42,3	42,4
6-10	152	41,3	83,7
11-20	43	11,7	95,4
21-50	17	4,6	100,0
Total	368	100,0	

The size of the teams appears to be mostly smaller than 10 persons. Only less than five percent of the organizations have teams with more than 20 persons, and none of the organizations have teams bigger than fifty persons.

4.2 Usage of Agile and Lean Methods

Usage of agile methods in the respondents organizational unit was reported by 55% (n=225) and usage of lean methods by 24% (n=99) of the respondents. The usage of agile or lean methods or a combination of these, are presented in Table 6.

Table 6. Usage of Agile and Lean Methods

Agile and Lean usage	n	%
Only Agile	137	33,6
Agile and Lean	88	21,6
Only Lean	11	2,7
No Agile or Lean	172	42,2
Total	408	100,0

The majority of respondents' organizational units (58%) appears to be using agile and/or lean methods. Agile seems still to be more popular than lean, although they are also often used in

combination (37% of agile and/or lean users). Only 2,7% of the respondents reported that they are using only lean methods. Earlier studies on the level of agile usage reported varying results, from 80% of respondents using agile in the survey conducted by Version One [24] to 32% in the survey conducted at Microsoft [7]. While the results of this study do not indicate as high level of agile usage as Version One, they reveal the strong position that agile methods are taking in software development. It is also significant that 24% of respondents reported to be using lean, when earlier studies have reported much lower levels of lean usage (2% by both [24] and [26]).

The time how long agile and lean methods have been used in the respondents' organizational units is presented in Table 7.

Table 7. Duration of Using Agile and Lean Methods

Usage time	Agile methods		Lean methods	
	n	%	n	%
Less than 1 year	32	14,2	22	22,2
1-2 years	78	34,7	50	50,5
2-5 years	99	44,0	18	18,2
More than 5 years	16	7,1	9	9,1
Total	225	100,0	99	100,0

Based on the data, agile methods seem to be used longer time than lean methods, as agile methods Median and Mode class is "2-5 years" and lean methods "1-2 years". According to the data 51% (n=115) have been using agile more than two years, in comparison to 27% (n=27) of lean usage more than two years.

There appeared also difference in the level of adoption of agile and lean methods. According the data, agile methods are used by all of the teams in 26% and by only some of the teams in 74% of the organizational units. The corresponding percentages of lean methods usage are 14% and 86%. These results show that agile methods are more widely spread in the organizations than lean.

4.3 Usage of specific methods, practice and principles

As a part of the survey, the popularity of different agile and lean methods, as well as agile and lean practices and principles were studied. The usage of specific agile methods, presented in Table 8, was measured by requesting the respondents to indicate which of the methods were in use in their organizational unit.

Table 8. Usage of specific agile methods

Methods	n	%
Scrum	196	83,1
Extreme Programming (XP)	43	18,1
Agile Modeling	27	11,4
Feature-Driven Development (FDD)	21	8,9
Kanban	11	4,7
Adaptive Software Development	10	4,2
Dynamic Systems Development Method (DSDM)	6	2,5
TDD	4	1,7
Crystal Methods	2	0,8
Other	18	7,6

Salo and Abrahamsson [27] reported that the respondents of their survey, conducted in 2008, were more familiarized with XP than with Scrum. The results of our study show that Scrum is clearly

the most widely used method. These results could reveal a trend towards Scrum to the detriment of XP. It is remarkable that although kanban and TDD were not preliminary included as agile methods (kanban is usually associated with lean and TDD is more considered as a practice of XP than a method itself), both were identified as agile methods by the respondents in the open field that was included for collecting other methods not included as default. Otherwise, the “Other” methods were most typically in-house methods or modifications of well-known methods that were adapted to the organization’s specific needs and practices.

The application of practices and principles were measured by five point scale, asking the respondents to indicate how frequently they were applied them in their organizational unit (value 1 indicating never and 5 indicating systematically). These results are reported correspondingly in Tables 9 and 10, where mean and mode presents the average frequency of application of the particular practice or principle.

Table 9. Usage of specific agile practices

Practices	n	Mean	Median
Prioritized work list	204	4,2	4
Iteration/sprint planning	203	4,1	4
Daily stand-up meetings	209	3,7	4
Unit testing	199	3,7	4
Release planning	196	3,9	4
Active customer participation	196	3,5	4
Self-organizing teams	194	3,5	4
Frequent and incremental delivery of working software	189	4,1	4
Automated builds	185	3,5	4
Continuous integration	182	3,8	4
Test-driven development (TDD)	179	2,7	3
Retrospectives	177	3,6	4
Burn-down charts	174	3,2	3
Pair programming	174	2,4	2
Refactoring	163	3,4	3
Collective code ownership	159	3,3	3
Other	9	1,8	1

Regarding to the agile practices in use, the results are well aligned with the results of earlier studies. However, it is significant that most of the practices are in general used without big differences in their usage, oppositely to the results of previous studies that reported non very much usage of practices such as TDD [7, 26, 27] and pair programming [7, 27].

Lean principles, presented in Table 10, were asked also from those respondents applying only agile, since they may be using them even if they are not aware of these principles belongs to lean.

Table 10. Usage of specific lean principles

Principles	n	Mean	Median
Focus on creating customer value	209	3,9	4
Eliminate waste and excess activities	199	3,4	3
Create a culture of continuous improvement	198	3,5	4
Do it right the first time	196	3,4	3
Respect and empower people	190	3,8	4
Minimize inventory or work in progress	189	3,2	3
Pull from demand	184	3,6	4

Focus on optimizing the whole system and not only local optimizations	181	3,3	3
Continuous flow of small batches in the development process	179	3,5	4
Make decisions as late as possible	175	3,0	3
Root source analysis is done after problems are discovered	174	3,2	3
Look simultaneously for multiple solutions	174	3,2	3
Create trusted relationships with suppliers	165	3,6	4
Create cadence	147	2,9	3

4.4 Goals, Challenges and Effects in adoption

The agile and lean users were also requested to identify their organisations’ goals, challenges and experienced limitations in adopting agile and lean, as well as effect of the adoption in their organisational units. The goals were measured by requesting the respondents to indicate the specific goals in adoption and the results are presented in the Table 11. The challenges and affecting limitations were measured by using five point scale, asking the respondents to rate how significant the particular challenge or limitation has been and the results are presented in the Tables 12 and 13. Mean and median in these tables presents the average significance of the particular challenge or limitation in the adoption. The effects of adoption to the organisational units were measured also with five point scale indicating the level of improvement. The result of the organisational effects are presented in the Table 14.

Table 11. Goals in agile and lean adoption

Goals	n	%
To increase productivity	158	66,9
To improve product and service quality	145	61,4
To reduce development cycle times and time-to-market	137	58,1
To improve process quality	113	47,9
To increase the ability to adapt to changes in the business environment	110	46,6
To improve team communication	100	42,4
To improve development flow	99	41,9
To reduce risks	86	36,4
To remove waste and excess activities	75	31,8
To decrease development costs	75	31,8
To improve customer understanding	65	27,5
To create transparency within the organization	63	26,7
To improve stakeholders' satisfaction	57	24,2
To improve organizational learning	45	19,1
To improve the management of business/product value	42	17,8
To establish team-wide project comprehension	33	14,0
To improve our understanding of the whole value stream	31	13,1
To achieve success others have achieved using lean methods	27	11,4
Other	6	2,5

The results in the Table 11 show the order of importance of different goals in agile and lean adoption. To increase the productivity and quality of the products/services as well as to reduce time-to-market are reported as the main goals in organizational decisions to use agile and lean methods. Moreover, to increase the ability to adapt to changes in the business environment, that has been reported in earlier studies as a key goal for adopting agile [8, 24] and should be essential in agile methods by definition [29], appears also in the top positions of our ranking.

Table 12. Challenges in agile and lean adoption

Challenges	n	Mean	Median
Top management commitment	201	4,0	4
Customer/supplier collaboration	192	3,9	4
Cultural change/translating agile/lean principles from development teams to the rest of the business	190	3,8	4
Measuring agile/lean success	190	3,6	4
Resistance to change	16	3,6	4
Defining business value	184	3,9	4
Need for specialized skills	183	3,1	3
Tailoring agile/lean practices	182	3,5	4
Lack of formal guidelines	176	2,8	3
Inadequate documentation	175	3,2	3
Scalability of agile/lean methods	174	3,6	4
Inadequate training	174	3,3	3
Synchronizing activities	172	3,4	4
Synchronizing activities	168	3,5	4
Loss of management control	168	2,8	3
Lack of big design up front	167	3,0	3
Fixed price contracts	161	3,3	3
Steep learning curve	159	3,1	3
Inappropriateness of existing technologies/tools	158	3,1	3
Achieving flow	157	3,5	4
Decreased predictability	155	3,1	3
Other	12	3,2	3

The results in the Table 12 show the order of significance of different challenges in agile and lean adoption. Top management commitment appears as the top challenge for adopting agile and lean methods. This result is opposed to the findings of the study by Chow and Cao [19], which rejected the strong management commitment as a critical success factor in agile software projects. Customer/suppliers collaboration is the second most important challenge that software organizations have for adopting agile and lean methods. Customer collaboration and commitment have been also identified as critical success factors in [19, 21].

Table 13. Limitations in agile and lean adoption

Limitations	n	Mean	Median
Limited support for developing large, complex software	189	3,3	3
Limited support for development involving large teams	185	3,1	3
Limited support for sub-contracting	181	2,8	3

Limited support for development involving legacy systems	179	3,1	3
Limited support for distributed development environments	178	2,9	3
Limited support for building reusable artifacts	176	3,0	3
Limited support for developing safety/mission-critical software	174	2,9	3
Other	16	3,3	3

The results in the Table 13 show the order of significance of different limitations in agile and lean adoption. One of the most controversial topics in the usage of agile methods is its ability to work in distributed environments. While [7, 8, 9, 25] have found that scaling agile to distributed environments could be a barrier for agile methods, [21] did not find statistical significance to assert that team distribution is a factor that impact in adopting agile. The results of our survey are aligned with those considering agile as a limited approach in distributed settings, since limited support for development involving large teams and limiting support for distributed development environment appear as important limitations in opinion of respondents of the survey.

Table 14. Effects of adoption of agile and lean

Effect	n	Mean	Median
Improved team communication	204	4,0	4
Enhanced ability to adapt to changes	203	3,9	4
Increased productivity	201	3,8	4
Enhanced process quality	198	3,7	4
Improved learning and knowledge creation	197	3,7	4
Enhanced software quality	196	3,8	4
Accelerated time-to-market/cycle time	192	3,7	4
Reduced waste and excess activities	190	3,5	4
Improved customer collaboration	190	3,7	4
Improved organizational transparency	187	3,5	4
Improved customer understanding	188	3,7	4
Reduced risks	184	3,4	3
Improved alignment between IT and business objectives	180	3,4	3
Enhanced value creation	178	3,6	4
Improved stakeholder satisfaction	169	3,6	4
Reduced costs	163	3,2	3
Other	13	3,8	4

According to the results of the survey, respondents applying agile and lean reported benefits in almost of the aspects that theoretically should be improved using agile and lean methods. Improved communication, as already claimed by [4], increased ability to adapt to changes and increased productivity were the

three most common benefits experienced by the respondents of the survey.

4.5 Non-adopter's reasons and plans to adopt agile and lean methods

The survey responses included also 137 practitioners in whose organizational units agile or lean methods were not in use. Those non-adopters were presented questions related to the reasons why they did not consider these methods appropriated for their software development activities. The results of these reasons are presented in Table 15.

Table 15. Reasons for not adopting agile and lean

Reasons for not adopting	n	%
Lack of knowledge and training	64	46,7
Too traditional organizational culture	59	43,1
Lack of support or commitment from the management	28	20,4
Fixed price contracts	25	18,2
Customers are not ready for agile/lean methods	22	16,1
Resistance to change	18	13,1
Inappropriate technology and tools	17	12,4
Incompatible business domain (please specify your business domain)	12	8,8
The burden of changing to agile/lean methods	10	7,3
Lack of quality assurance procedures	10	7,3
Unstable project requirements	9	6,6
Lack of progress-tracking mechanism	9	6,6
Our organization lacks customer understanding	7	5,1
Lack of scalability	6	4,4
Lack of support for reward system	6	4,4
Lack of big design up front	6	4,4
Limited support for building reusable artifacts	3	2,2
Limited support for distributed development environments	3	2,2
Decreasing predictability	2	1,5
Other	27	19,7

The results in Table 15 reveal that clearly the most common reasons preventing agile and lean adoption are lack of knowledge and training and organizational culture.

The future adoption plans were also charted, and those results are presented in the Table 16.

Table 16. Plans to adopt agile and lean in the future

Planned adoption	n	%
Within a year	14	10,2
Within two years	8	5,8
Not planned	52	38,0
Don't know	63	46,0
Total	137	100,0

Based on the result in Table 16, it seems that only a very few, organizational units of 22 respondents, have clear plans to adopt agile and lean in the near future. Majority of the organizations

have not been thinking or planned yet utilization of agile and/or lean methods.

The agile and lean methods planned to be adopted in the future by those respondents that have future adoption plans, are presented in the Table 17.

Table 17. Agile and lean methods planned to be adopted

Planned methods ⁶	n	%
Scrum	19	90
Lean Software Development	3	14
Agile Modeling	2	10
Extreme Programming (XP)	1	5
Dynamic Systems Development Method (DSDM)	0	0
Feature Driven Development (FDD)	0	0
Crystal Methods	0	0
Adaptive Software Development	0	0
Other	3	14

The reasons of those who are planning to adopt agile and lean methods are presented in the Table 18.

Table 18. Reason for adoptin agile and lean in the future

Reasons for planned adoption	n	%
To increase productivity	14	64
To decrease development cycle times and time-to-market	10	45
To improve development flow	10	45
To improve product or service quality	9	41
To reduce risks	9	41
To improve process quality	8	36
To decrease development costs	8	36
To increase ability to adapt to changes in the business environment	8	36
To reduce waste and excess activities	6	27
To improve organizational learning	6	27
To improve stakeholders' satisfaction	6	27
To increase team communication	6	27
To improve customer understanding	4	18
To create transparency within the organization	4	18
To establish team-wide project comprehension	4	18
To improve the management of business/product value	3	14
To improve understanding of the whole value stream	3	14
To achieve success others have achieved using agile/lean methods	1	5
Other	1	5

Similarly to the reason of those respondents already using agile and lean methods, from the results of the Table 18, it can be seen that increasing the productivity and reducing development time are the most important reasons for considering agile and lean methods usage in future plans.

⁶ Respondents were able to select multiple options

5. CONCLUSIONS AND LIMITATIONS OF THE STUDY

The goal of this study is to investigate the adoption and usage of agile and lean methods in the software industry. Using a large-scale exploratory survey, the study aims to analyze (i) the current state of agile and lean adoption and usage in the software industry, (ii) the usage of specific agile and lean methods, practices and principles and (iii) the goals, challenges and effects when agile and lean methods are adopted. Furthermore, the study investigates (iv) the reasons for not adopting agile and lean methods as well as the future plans of organizations that are not currently using them.

The results of the study are based on more than four hundred responses collected by a web-survey conducted in the Finnish software industry in 2011. Given the reported performance of Finnish IT industry (e.g. second highest productivity country in the world according to The Economist Intelligence Unit [23] and the fact that large number of practitioners taking part in the survey belonged to multinational companies with premises in several countries, the conclusions can have interest to most of the existing software development communities.

The results of the study indicate that: (i) Most of respondents' organizational units were using agile and/or lean methods (58%). Moreover, lean appears to be used by 24% of respondents, mainly in combination with agile (21%); (ii) Most of agile practices are generally used without big differences in the usage between practices. Prioritized work list was reported as the most used agile practice and collective code ownership as the least used. Although lean principles appear to be less used than agile practices, principles such as eliminate waste and excess activities and focus on creating customer value are quite used by the respondents of the survey. (iii) To increase the productivity and quality of the products/services as well as to reduce time-to-market are reported as the main goals for adopting agile and lean methods. Furthermore, they are also reported as important benefits of the usage of agile and lean methods. Regarding to challenges and limitations, to get strong management commitments seem to be challenging organizations using agile and lean methods. The ability to work in distributed environments has also been reported as one of the main limitations of the usage of agile and lean methods. (iv) Finally, the most common reasons preventing agile and lean adoption are lack of knowledge and training and working in organizations with too traditional culture. Moreover, most of the non-adopter respondents have not been thinking or planned yet utilization of agile and/or lean methods in the near future.

As the survey included quite a large sample of software professionals in Finland, covered by well representative registry of FIPA, it provides a good descriptive view to the state of agile and lean usage and adoption in the software industry. However, the study has some limitations that should be taken into account when interpreting the results. At first, the study subjects were individual persons who represented different organizations. Although in order to study agile and lean usage more exactly in organizational level, it would be more appropriated to use organizations as sampling units, it has to be considered that organizations may have different units that may be at different levels of agile and lean usage. Therefore, it would have been impossible for a single person to answer for the whole company. Associated to this issue, the software professionals in the study were acting in varying positions and roles in the organizations. The persons in different organizational positions may have somehow different views to the organizational practices, as well

as varying knowledge about agile and lean methods, which might affect at some level to the reliability of the results. Furthermore, the inconsistency in how agile and lean are understood can impact the results of the study. Since definitions of agility and leanness are not clearly agreed and the study is explorative in nature, we did not want to restrict the usage of agile and/or lean to specific definitions. Thus, the survey was designed for whatever software practitioners understand to be using agile and/or lean is, starting with the questions "Are you currently applying agile (lean) methods in your organizational unit?" and going deeper with questions about specific methods, practices and principles that were in use. Finally, the sample included professionals from different types of organizations; additionally to software companies it included also a few educational and public organizations. Even if this can be seen as a richness of the study, limiting the research only to software companies might have given more exact view to the software business. Despite of its limitations, the study can be seen to provide valuable descriptive information about the contemporary state of agile and lean adoption in software developing organizations.

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